KDK College of Engineering, Nagpur.
Department of Artificial Intelligence and Data Science

First Project Title Seminar

on

HealthGrid AI
Privacy-Preserving Federated Machine Learning for Healthcare

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ABSTRACT

HealthGrid AI is a privacy-preserving Federated Learning system designed for collaborative Machine Learning in healthcare. Healthcare data is sensitive and is usually stored separately by different hospitals. Instead of collecting all patient records at one central location, the proposed system keeps the data inside each participating hospital. Each hospital trains the Machine Learning model locally using its own data and sends only the learned model update to a federated server. The server aggregates the updates to create an improved global model and sends the updated model back to the hospitals for the next training round. Thus, hospitals can contribute to a common AI model without directly sharing their raw patient records. For the academic prototype, public or synthetic healthcare data will be used to represent separate hospitals.

INTRODUCTION

- Healthcare institutions generate large amounts of patient and clinical data such as medical history, laboratory results and diagnosis records.
- Machine Learning can use this data to learn patterns and support healthcare prediction and decision-making.
- The main challenge is that healthcare data is sensitive and is stored separately in different hospitals.
- HealthGrid AI uses Federated Learning so hospitals can train a common model without sending raw patient records to a central server.
- Each hospital trains the model locally and sends only model updates to the federated server
- The server combines these updates to create a better global model and sends it back to the hospitals.

LITERATURE SURVEY

Sr. No.	Name of Article Reviewed (Research Paper/Other Media)	GAP's in the Given Article / Research Paper
1.	H. Brendan McMahan (2017) – Communication-Efficient Learning of Deep Networks from Decentralized Data	H. Brendan McMahan (2017) – Communication-Efficient Learning of Deep Networks from Decentralized Data
2.	Theodora S. Brisimi (2018) – Federated Learning of Predictive Models from Federated Electronic Health Records	Focused on EHR-based prediction; broader healthcare collaboration can be explored.
3.	Nicola Rieke (2020) – The Future of Digital Health with Federated Learning	Discussed digital-health applications; practical deployment still has challenges.
4.	Micah J. Sheller (2020) – Federated Learning in Medicine: Facilitating Multi-Institutional Collaborations Without Sharing Patient Data	Demonstrated medical collaboration; wider real-world deployment needs further work.
5.	Peter Kairouz (2021) – Advances and Open Problems in Federated Learning	Reviewed FL challenges and open problems; a complete healthcare prototype is not provided.

OBJECTIVES

- **Privacy :** Keep raw healthcare data inside the local hospital environment.
- **Collaborative Learning :** Allow multiple hospitals to train one common ML model.
- **Local Training :** Train the model at each hospital using only its local dataset.
- **Federated Server :** Receive model updates and combine them using FedAvg.
- **Global Model :** Send the improved model back to hospitals for the next round.
- **Prototype :** Demonstrate the complete process using public or synthetic healthcare data.

PROBLEM STATEMENT

Healthcare data is distributed across hospitals and contains sensitive patient information.

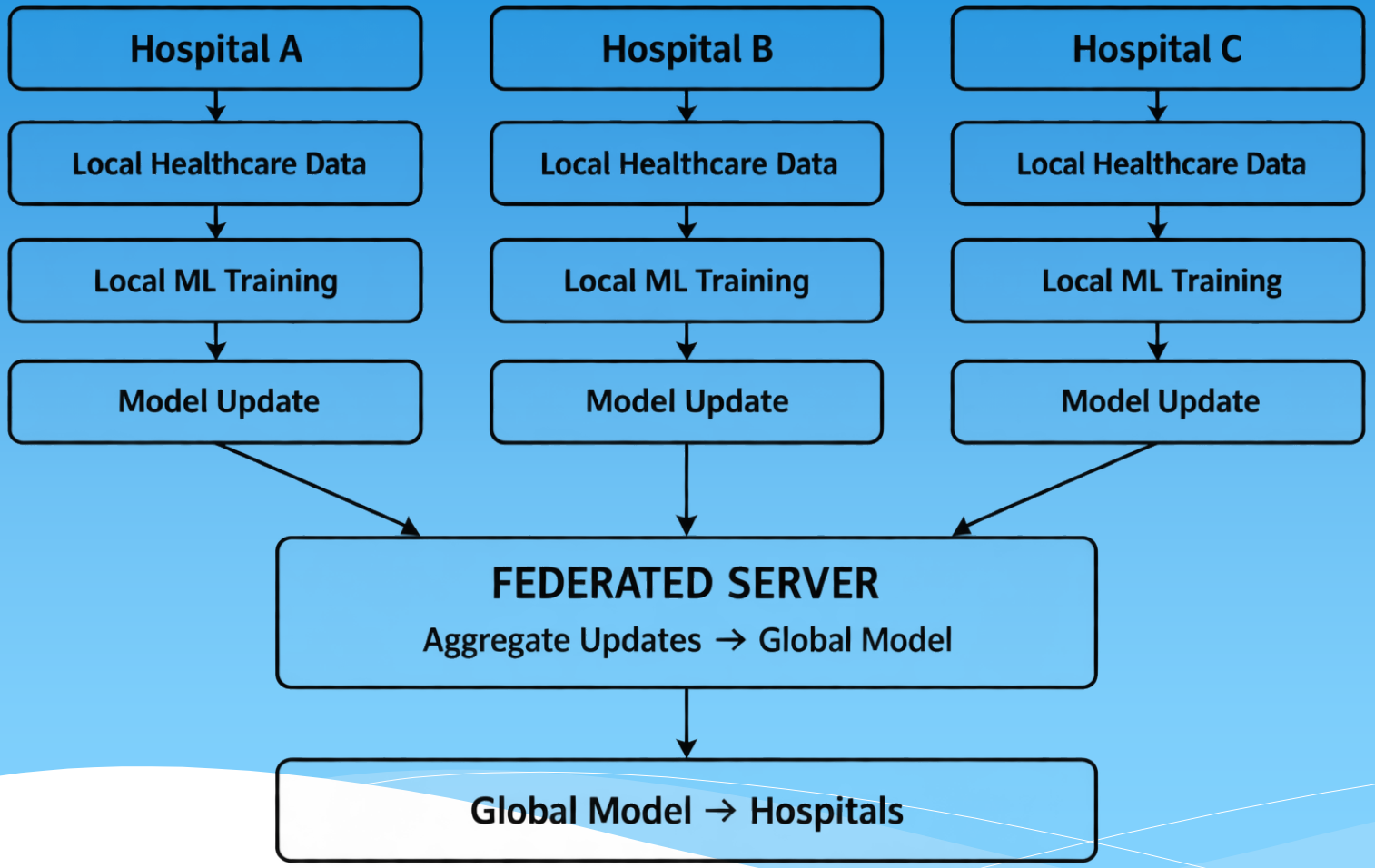
In a traditional centralized approach, hospitals would need to send their records to a common database. This creates privacy, security and data-sharing concerns.

CORE PROBLEM:

How can multiple hospitals collaboratively train a Machine Learning model without sending their raw patient records to a central location?

HealthGrid AI addresses this using Federated Learning: data stays local, training happens locally, and only model updates are shared with the federated server.

Flowchart


Hospital A
Local Healthcare Data
Local ML Training
Model Update
Hospital B
Local Healthcare Data
Local ML Training
Model Update
Hospital C
Local Healthcare Data
Local ML Training
Model Update
FEDERATED SERVER
Aggregate Updates → Global Model
Global Model → Hospitals

Methodology

- Data Preparation: Use public or synthetic healthcare data and divide it into separate datasets representing different hospitals.
- Local Client Setup: Create one federated client for each hospital so it can access only its own local data.
- Model Initialization: Create a common initial Machine Learning model and distribute it to all clients.
- Local Training: Each client trains the model locally without sending raw patient records to the server.
- Model Update: Each client sends only its trained model update to the federated server.
- Aggregation: The server combines the updates using Federated Averaging (FedAvg) to create the global model.
- Evaluation: Test the global model and repeat the training process for multiple rounds.

PLAN FOR FUTURE WORK

- Complete the working prototype using three simulated hospital clients.
- Test multiple Federated Averaging training rounds and compare model performance.
- Build a simple dashboard for training status, model performance and predictions.
- Study how different hospital data distributions affect the global model.
- Add privacy techniques such as secure aggregation and differential privacy.
- Explore authorized EHR/API integration for future real-world deployment.

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THANK YOU…